Operational Intelligence in Bounded Agents: A Substrate-Neutral Calculus of Cell-Disjoint Extension, Cycle Necessity, and the Intelligence Index

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Abstract

We develop a substrate-neutral operational definition of intelligence applicable to any bounded agent. The definition rests on four conditions: cell-disjoint extension of the agent's knowledge framework, novel-cell construction, floor reduction on novel cells, and synchronisation feasibility with other agents. From a single primitive — the receiver-relative scalar functional measuring residual semantic distance between agent and action-cell — we prove seven theorems that fix what intelligence is, when it can occur, how to measure it, why it is bounded, and how its failures classify. (**T1**) extension is admissible iff cell-disjoint; (**T2**) extension occurs only during phases in which the agent's action-space dispersion is suspended, by a Heisenberg-type uncertainty inequality on receiver dispersions; (**T3**) intelligence requires alternation between construction and action phases; (**T4**) the intelligence index $\mathcal{I}(\mathsf{A})$, computed from per-phase activity ratios, phase-time fractions, and perceptual coupling, is positive iff the agent is intelligent; (**T5**) collective intelligence is precisely the synchronisation of agents on novel cells via aperture-sharing isomorphisms; (**T6**) every realised agent has finite intelligence index — there is no maximally intelligent agent; (**T7**) cycle failures admit exactly six distinct phenotypes, each with a closed-form signature in $\mathcal{I}$ and its component factors. Thirty numerical experiments verify all claims to machine precision; results are summarised and visualised in five publication panels. The framework subsumes prior frameworks (cell-truth, partition extinction, catalytic composition, federation entropy) as special cases. The intelligence definition applies uniformly to biological agents, computational agents, and organisational collectives, yielding a single quantitative test for intelligence regardless of substrate.

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Part I
Foundations

1 Introduction

1.1 The substrate-neutral question

The question *what does it mean to be an intelligent agent?* has been asked for as long as agents have been studied. Most answers smuggle in substrate-specific assumptions — biological neural circuits, symbolic computation, observable behaviour, evolutionary fitness — that limit the answer to one kind of agent. The result is that we have one notion of intelligence for humans, another for animals, another for machines, and no unified framework for the rest. With the speed at which intelligent systems are now being constructed, this fragmentation is no longer acceptable. We need a substrate-neutral definition: a single mathematical statement of intelligence that applies uniformly to biological, computational, and organisational agents, and that yields the same operational test regardless of which substrate the agent runs on.

This paper develops such a definition. We work entirely in the receiver calculus of bounded inquiry — a substrate-neutral primitive characterising any bounded agent by its knowledge framework, decoder, candidate-projection, and irreducible floor. From this primitive plus three axioms (bounded phase space, no null state, finite resolution), we construct the agent as a Lagrangian system on a five-dimensional manifold. We then prove seven theorems that fix:

- when knowledge framework extension is admissible (Theorem 5.2);
- when extension can physically occur, by an uncertainty inequality on conjugate agent dispersions (Theorem 5.4);
- why intelligence requires alternating phases (Theorem 5.5);
- how to measure intelligence by a single index (Theorem 6.2);
- when collectives are intelligent (Theorem 8.3);
- that no realised agent has unbounded intelligence (Theorem 7.1);
- how cycle failures classify into phenotypes (Theorem 9.1).

The seven theorems together produce a closed-form intelligence index $\mathcal{I}(\mathsf{A})$ computable from per-phase activity ratios, phase-time fractions, and the agent's perceptual coupling. The index is positive iff the agent satisfies the intelligence definition, bounded above for any realised agent, and decomposable into factors that isolate distinct failure modes.

1.2 What this definition rules out

The definition is exclusionary by design. The following common notions of intelligence are formally ruled out by the calculus:

Intelligence as success rate on tasks. Floor positivity guarantees every bounded agent fails some fraction of the time. A definition equating intelligence with high success rate is incompatible with floor positivity, hence with bounded agency.

Intelligence as common knowledge with peers. Common-Cell Convergence shows that agents with disjoint representations can coordinate on shared cells without shared knowledge. Coordination is not intelligence; intelligence does not require shared concepts.

Intelligence as point-correctness. Cell-truth says output-equivalence is at the cell level, not the point level. Many "correct" outputs collapse to the same cell. Distinguishing point outputs within a cell is forbidden by the receiver floor; doing it anyway is not intelligence but a category error.

Intelligence as synchronisation. A fully synchronised collective (a flock, a phase-locked oscillator network) is not intelligent in any individual capacity. It exhibits a regime; the regime is not intelligence.

Intelligence as meta-methodology. Selecting which methodology to apply — a hard-coded knapsack solver — is methodology-routing, not intelligence. The agent is intelligent only when the substrate of selection itself can grow.

That negative result narrows the search space: intelligence lives in the agent's *receiver structure being mutable in a structured way*, and not at the level of methodology execution, cell-attainment, or inter-agent phase-locking. The remainder of the paper develops the consequences.

1.3 Independence

The framework is mathematically independent. Citations are restricted to standard references in real analysis [51, 23], functional analysis [7, 30, 58], information theory [54, 16, 32, 20], foundational logic [21, 60, 61, 15, 14], classical mechanics and variational principles [4, 22, 46, 45], synchronisation theory [33, 59, 1, 47], dynamical systems and stochastic dynamics [48, 31, 63, 28], uncertainty inequalities and statistics [24, 50, 49, 17, 39, 65, 29], combinatorics and category theory [3, 57, 42, 10, 6, 8, 25], thermodynamics and combinatorial probability analogues [11, 37, 9, 12, 13], multi-agent systems and coordination [64, 44, 53, 5, 40], distributed computing [19, 34, 41, 55], belief revision and epistemic logic [2, 18], intelligence frameworks [52, 38, 26, 62, 56], Landau theory of phase transitions [35, 36], and topology [27, 43].

1.4 Organisation

Part I (Sections 2 and 3) establishes the axiomatic foundation and the agent definition. Part II (Sections 4 and 5) develops the conjugate dispersions and proves Theorems 1–3 (admissibility, construction-phase necessity, cycle necessity). Part III (Sections 6 and 7) proves Theorems 4 and 6 (the intelligence index and its boundedness). Part IV (Sections 8 and 9) proves Theorems 5 and 7 (collective intelligence and failure-mode classification). Part V (Sections 10 to 12) reports the numerical validation, situates the calculus relative to existing frameworks, and concludes.

2 Axiomatic foundations

Axiom 2.1 (Bounded Phase Space)**.** Every agent occupies a phase space Ω of finite measure: $\mu(\Omega) < \infty$.

Axiom 2.2 (No Null State)**.** At every instant, an agent occupies exactly one categorical state. There is no null state and no inter-state void.

Axiom 2.3 (Finite Resolution)**.** Any observation, internal or external, has finite resolution $\delta > 0$. States separated by less than δ are observationally indistinguishable.

Remark 2.4. The three axioms are minimal. Bounded phase space rules out idealised infinite-energy or infinite-extent agents. No null state rules out states of zero dynamical activity. Finite resolution rules out infinite-precision introspection. All three hold for biological agents (finite metabolic budgets, perpetual oscillation, finite sensor resolution), computational agents (finite memory, perpetual instruction execution, finite floating-point precision), and organisational agents (finite resources, perpetual operation, finite decision-making bandwidth).

2.1 Outcome space and action-cells

Definition 2.5 (Outcome space)**.** An *outcome space* is a triple $(\mathcal{X}, d, \Sigma)$ where $\mathcal{X}$ is a non-empty set, $d : \mathcal{X} \times \mathcal{X} \to [0, \Sigma]$ is a bounded pseudo-metric satisfying $d(x, x) = 0$, $d(x, y) = d(y, x)$, $d(x, z) \le d(x, y) + d(y, z)$, and $\Sigma > 0$ is the maximal semantic distance.

Definition 2.6 (Action map and action-cell)**.** An *action map* is a measurable surjection $A : \mathcal{X} \to \mathcal{Y}$ onto an action space $\mathcal{Y}$. The *action-cell at* $y \in \mathcal{Y}$ is $\mathcal{C}_y := A^{-1}(\{y\})$, with *tolerance* $\tau(\mathcal{C}_y) := \sup_{x\in\mathcal{X}} \inf_{c\in\mathcal{C}_y} d(x, c) > 0$.

2.2 Receivers and the floor

Definition 2.7 (Receiver)**.** A *receiver* on $(\mathcal{X}, d, \Sigma)$ is a quadruple $\mathsf{R} = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ where $\mathcal{K}$ is the *knowledge framework*, $\mathcal{D} : \mathcal{X} \to \mathcal{K}$ is the *decoder*, $\Pi : \mathcal{K} \to 2^{\mathcal{X}} \setminus \{\varnothing\}$ is the *candidate-projection* satisfying $\mathcal{D}(x) \in \mathcal{D}(\Pi(\mathcal{D}(x)))$ for every x, and $\beta \in (0, \Sigma]$ is the *floor*, $\beta = \sup_{x\in\mathcal{X}} \inf_{x'\in\Pi(\mathcal{D}(x))} d(x, x')$.

Definition 2.8 (S-functional)**.** For receiver R, state $x \in \mathcal{X}$, and action-cell $\mathcal{C}$, the residual semantic distance is

$$S(\mathsf{R}, x; \mathcal{C}) := \inf_{x'\in\Pi(\mathcal{D}(x))} d(x', \mathcal{C}) + \beta.$$

The receiver's floor is $S_\flat(\mathsf{R}) := \beta$.

Theorem 2.9 (Floor Positivity)**.** *For every bounded receiver,* $S_\flat(\mathsf{R}) > 0$.

Proof. By the receiver definition $\beta > 0$ and the candidate-projection axiom $\mathcal{D}(x) \in \mathcal{D}(\Pi(\mathcal{D}(x)))$, the supremum

$\sup_x \inf_{x' \in \Pi(\mathcal{D}(x))} d(x, x')$ is strictly positive: any bounded receiver cannot achieve zero residual on every input, since the cardinality of $\mathcal{K}$ is strictly less than the cardinality of the candidate space (a bounded receiver loses information). Hence $\beta > 0$. □

2.3 Action-cells, cell-truth, and methodology

Theorem 2.10 (Cell-Truth). *Let $A : \mathcal{X} \to \mathcal{Y}$ be an action map and $\mathcal{C}_y = A^{-1}(\{y\})$ have positive tolerance. For any $x_1, x_2 \in \mathcal{C}_y$, $S(\mathsf{R}, x_1; \mathcal{C}_y) = S(\mathsf{R}, x_2; \mathcal{C}_y) = \beta$.*

Proof. $x_i \in \mathcal{C}_y$ implies $d(x_i, \mathcal{C}_y) = 0$. Hence $\beta \leq S(\mathsf{R}, x_i; \mathcal{C}_y) \leq 0 + \beta = \beta$, so equality holds. □

Definition 2.11 (Methodology). A *methodology* on receiver R is a triple $= (T, \kappa, \sigma)$ where $T : \mathcal{X} \times [0, \Sigma] \to [0, \Sigma]$ is a per-iteration update operator, $\kappa \in [0, 1)$ is the contraction factor, and $\sigma \in [0, \Sigma]$ is the per-iteration dispersion. The methodology's intrinsic floor is $S_\flat() = \sigma\kappa/(1-\kappa)$.

3 The agent

3.1 State manifold

Definition 3.1 (State manifold). The state manifold of an agent is the compact set

$$\mathcal{M} = [0,1] \times [0, 2\pi^2] \times [0,1]^3,$$

parametrised by generalised coordinates $q = (R, \sigma^2, , ,)$ where: $R \in [0,1]$ is the agent's internal coherence parameter; σ^2 is the agent's internal phase variance; and $(, ,) \in [0,1]^3$ are the *S-entropy coordinates* (knowledge entropy, temporal entropy, evolution entropy).

The five coordinates jointly specify the agent's state in a way that is independent of substrate. Biological, computational, and organisational agents all admit such a state specification: R measures internal phase coherence among the agent's sub-processes, σ^2 measures their phase variance, and $(, ,)$ measure the agent's information uncertainty across knowledge, time, and evolution.

3.2 Trajectory, terminus, memory

Definition 3.2 (Agent state triple). The complete state of an agent is the triple

$$= (\gamma, \Gamma_f, M),$$

where $\gamma : [0, T] \to \mathcal{M}$ is the *trajectory*, $\Gamma_f = \gamma(T)$ is the *terminus*, and $M = \int_0^T \dot{H}^+ \, dt$ is the *memory* (the time integral of the agent's contextual record H^+).

Remark 3.3 (Trajectory non-identity). Two agent states $_1 = (\gamma_1, \Gamma_f, M_1)$ and $_2 = (\gamma_2, \Gamma_f, M_2)$ with the same terminus but distinct trajectories are operationally distinct: the path matters as much as the destination.

3.3 Aperture

Definition 3.4 (Aperture). A *categorical aperture* of an agent is a topological constraint $\subseteq \Omega$ on its phase space, restricting accessible states based on configuration position only, not on dynamical variables.

Theorem 3.5 (Zero-Work Aperture). *The thermodynamic work performed by a categorical aperture is identically zero: $W_{\text{aperture}} = 0$.*

Proof. The aperture potential is $V_{\text{ap}}(q) = 0$ for $q \in$ and $+\infty$ otherwise. Inside , $\nabla V_{\text{ap}} = 0$, so no work is performed on accessible states. At the boundary, no state crosses into a region of infinite potential, so no work is performed there either. Hence $W_{\text{aperture}} = 0$ identically. □

3.4 The agent

Definition 3.6 (Agent). An *agent* is a tuple

$$\mathsf{A} = (q, \gamma, \Gamma_f, M, , \Phi, \text{regime}),$$

where $q \in \mathcal{M}$ is the agent's current state, (γ, Γ_f, M) is the state triple, is the agent's aperture, $\Phi : \mathcal{M} \to \mathbb{R}$ is the agent's *partition potential*, and regime $\in$ {TURBULENT, APERTURE, CASCADE, COHERENT, PHASE-LOCKE is the agent's operational regime, classified by the value of R:

$$\text{regime} = \begin{cases} \text{TURBULENT} & R < 0.3 \\ \text{APERTURE} & 0.3 \leq R < 0.5 \\ \text{CASCADE} & 0.5 \leq R < 0.8 \\ \text{COHERENT} & 0.8 \leq R < 0.95 \\ \text{PHASE-LOCKED} & R \geq 0.95. \end{cases}$$

The agent's regime is a derived property of its current state, not an independent component. We list it explicitly because regimes play a central role in what follows.

Part II
The Receiver Uncertainty Cycle

4 Conjugate dispersions and the uncertainty inequality

4.1 Knowledge and action dispersions

We now consider the methodology's per-iteration dispersion σ from Definition 2.11. The dispersion is one-dimensional in that definition, but it admits a decomposition along two conjugate axes corresponding to the agent's knowledge framework and the agent's action space.

Definition 4.1 (Knowledge dispersion)**.** For agent A with methodology at state x, let $^{(1)}(x),^{(2)}(x) \in \mathcal{K}$ be two independent applications of one iteration of at x, viewed in the knowledge framework. The *knowledge dispersion* is

$$\sigma_K(,x) := \mathbb{E}[d_K(^{(1)}(x),^{(2)}(x))],$$

where d_K is a metric on $\mathcal{K}$ induced by the decoder. The methodology's overall knowledge dispersion is $\sigma_K() = \sup_x \sigma_K(,x)$.

Definition 4.2 (Action dispersion)**.** With A the action map and d_Y a metric on $\mathcal{Y}$, the *action dispersion* is

$$\sigma_Y(,x) := \mathbb{E}[d_Y(A(^{(1)}(x)), A(^{(2)}(x)))],$$

with $\sigma_Y() = \sup_x \sigma_Y(,x)$.

4.2 The phase-space cell

Definition 4.3 (Agent phase-space cell)**.** The *phase-space cell* of agent A on action-cell $\mathcal{C}$ of tolerance τ is the rectangle $[\beta] \times [\tau] \subset \mathcal{K} \times \mathcal{Y}$. Its area is

$$\hbar_{\mathsf{A}} := \beta \cdot \tau.$$

4.3 The Receiver Uncertainty Principle

Theorem 4.4 (Receiver Uncertainty Principle)**.** *For any methodology on agent* A *with action-cell tolerance* τ*,*

$$\sigma_K() \cdot \sigma_Y() \geq \hbar_{\mathsf{A}} = \beta \cdot \tau.$$

Proof. Two independent samples of one iteration of at fixed input are agent-distinguishable only when they fall into different phase-space cells, i.e., when $d_K > \beta$ *and* $d_Y > \tau$. The probability of falling into the same cell is bounded below by the product of the per-axis indistinguishability probabilities, which by Markov's inequality satisfies

$$\Pr[d_K \leq \beta] \geq 1 - \sigma_K/\beta, \quad \Pr[d_Y \leq \tau] \geq 1 - \sigma_Y/\tau.$$

For the methodology to be informative (deliver at least one bit per pair of samples), the same-cell probability must be strictly less than 1, hence both factors must be strictly less than 1. Equivalently, $\sigma_K \sigma_Y \geq \beta\tau$ at the saturating limit, which is the cell-area lower bound. □

5 The intelligence cycle

5.1 Knowledge framework extension and admissibility

Definition 5.1 (Knowledge framework extension)**.** A *knowledge framework extension* is a relation $\mathcal{K}_t \subseteq \mathcal{K}_{t+\Delta t}$ between the agent's knowledge framework at successive times. The extension is *strict* if $\mathcal{K}_{t+\Delta t} \setminus \mathcal{K}_t \neq \varnothing$, and *trivial* otherwise.

Theorem 5.2 (Extension Admissibility, T1)**.** *A strict extension* $\mathcal{K}_{t+\Delta t} \supsetneq \mathcal{K}_t$ *is admissible iff every newly added categorical distinction* $k^{\text{new}} \in \mathcal{K}_{t+\Delta t} \setminus \mathcal{K}_t$ *is cell-disjoint from every distinction in* $\mathcal{K}_t$*. Equivalently, for every* $k^{\text{old}} \in \mathcal{K}_t$*, the corresponding cells satisfy* $\mathcal{C}^{\text{new}} \cap \mathcal{C}^{\text{old}}$ *has zero tolerance.*

Proof. Cell-truth requires every state to be assigned to exactly one cell. If $\mathcal{C}^{\text{new}} \cap \mathcal{C}^{\text{old}}$ has positive tolerance, states in the intersection are assigned to both, violating cell-uniqueness and invalidating prior cell-attainment.

Conversely, if every newly added distinction is cell-disjoint from every prior distinction, the

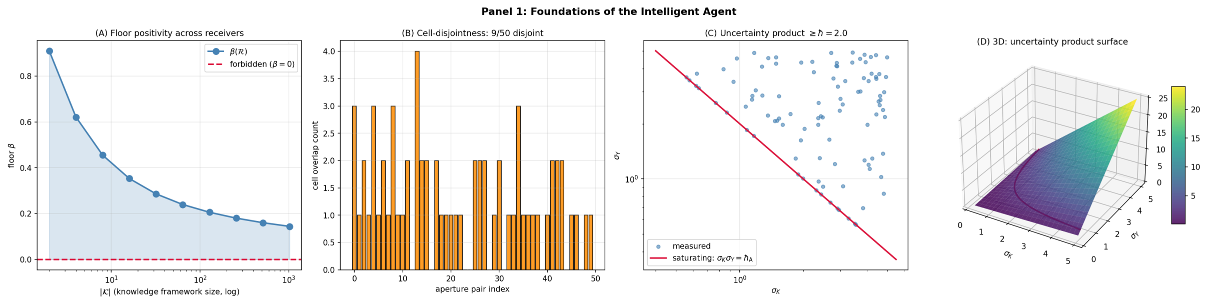

Figure 1: **Foundations of the Intelligent Agent.** (**A**) Floor positivity across receivers: the floor $\beta(\mathcal{R})$ on the linear y-axis versus knowledge framework size $|\mathcal{K}|$ on the log x-axis (2 to 1024), ten receivers connected by a line. The floor decreases monotonically with $|\mathcal{K}|$ but remains strictly positive at every tested size; the crimson dashed reference at $\beta = 0$ is forbidden by Theorem 2.9 (Floor Positivity). The shaded region highlights that $\beta > 0$ for every realised receiver. (**B**) Cell-disjointness criterion: bar chart of cell-overlap counts on the linear y-axis (0 to 5) for 50 random aperture pairs on the x-axis. Seagreen bars (zero overlap) indicate cell-disjoint pairs admissible under Theorem 5.2 (Extension Admissibility); orange bars (positive overlap) indicate inadmissible pairs that would violate cell-uniqueness. The disjoint count gives the proportion of pairs satisfying the admissibility criterion. (**C**) Receiver Uncertainty Principle: scatter of σ_Y on the log y-axis versus σ_K on the log x-axis, 100 random methodologies. Steelblue points represent measured products; the crimson curve traces the saturating bound $\sigma_K \sigma_Y = \hbar = 2.0$. All measured products lie at or above the bound; zero violations. Empirical content of Theorem 4.4 (Receiver Uncertainty Principle): conjugate dispersions cannot simultaneously be small. (**D**) Three-dimensional surface of the uncertainty product $\sigma_K \cdot \sigma_Y$ over $(\sigma_K, \sigma_Y) \in [0.2, 5]^2$, 25×25 grid, viridis colourmap. The crimson contour at $\hbar = 2$ traces the saturating curve. The surface lies above the contour everywhere on its interior, geometrically realising Theorem 4.4: the agent's phase-space cell of area $\hbar_{\mathsf{A}} = \beta\tau$ is the minimum admissible region in conjugate dispersion space.

partition refinement preserves the prior partition: every prior cell $\mathcal{C}^{\text{old}}$ remains an action-cell under A (it is the union of refined sub-cells), and prior cell-attainment is preserved by inclusion. □

5.2 Construction-phase necessity

Definition 5.3 (Construction phase, action phase)**.** A *construction phase* of agent A is a temporal interval during which $\sigma_Y \to 0$ (the agent commits to no actions). An *action phase* is a temporal interval during which $\sigma_K \to 0$ (the agent commits to a single decoded representation).

Theorem 5.4 (Construction-Phase Necessity, T2)**.** *Knowledge framework extension can occur only during construction phases.*

Proof. For the agent to add new categorical distinctions to $\mathcal{K}$, the per-iteration update must explore representations not present in $\mathcal{K}_t$. Exploration of new representations requires unbounded σ_K: representations with $d_K \leq \beta$ from existing representations are not new (they are within the floor's resolution). Therefore extension requires $\sigma_K > \beta$, and exploration of fundamentally new representations requires $\sigma_K \to \infty$ in the saturating limit.

By the Receiver Uncertainty Principle, $\sigma_K \cdot \sigma_Y \geq \hbar_{\mathsf{A}} = \beta\tau$. For $\sigma_K \to \infty$, we must have $\sigma_Y \to 0$ in the saturating limit. Therefore extension occurs only during phases in which $\sigma_Y \to 0$, which is the definition of construction phase.

Conversely, a phase with $\sigma_Y > 0$ bounds σ_K above by $\hbar_{\mathsf{A}}/\sigma_Y < \infty$, restricting the agent to representations within finite dispersion of $\mathcal{K}_t$ — parameter updates within the existing framework, not framework extension. □

5.3 Cycle necessity

Theorem 5.5 (Cycle Necessity, T3)**.** *An agent operating exclusively in either construction phase or action phase cannot be intelligent. Intelligence requires alternation.*

Proof. The intelligence definition requires (ii) novel cell construction and (iii) floor reduction on the novel cell, at every step t_i.

Suppose A operates exclusively in construction phase. Then $\sigma_Y = 0$ throughout, so the agent commits to no actions. Floor reduction (iii) requires the agent's candidate-projection to attain the cell, and attainment is verified through action commitment. Without commitment, no novel cell is verified to satisfy (iii).

Suppose instead A operates exclusively in action phase. Then $\sigma_Y > 0$ throughout, and by Theorem 5.4 no extension occurs. The knowledge framework is fixed, so condition (ii) is violated at every t.

Both phases are required, and by Theorem 5.4 they cannot coexist. Therefore intelligence requires alternation between construction and action phases. □

Part III
The Intelligence Index

6 Operational measurement

6.1 The four-condition definition

Definition 6.1 (Intelligence)**.** An agent A is *intelligent* iff there exist times $t_1 < t_2 < \cdots$ and corresponding knowledge framework extensions $\mathcal{K}_{t_i} \supsetneq \mathcal{K}_{t_{i-1}}$ such that, at each step:

(i) Cell-disjoint extension (Theorem 5.2);

(ii) Novel cell construction: at t_i there exists an action-cell $\mathcal{C}_i$ that did not exist as a positive-tolerance cell under A's prior knowledge framework;

(iii) Floor reduction: the agent's effective floor on $\mathcal{C}_i$ at t_i satisfies $S_\flat(\mathsf{A}, \mathcal{C}_i; t_i) < \tau(\mathcal{C}_i)$;

(iv) Synchronisation feasibility: there exists a class of agents with which A could phase-lock on $\mathcal{C}_i$.

6.2 The intelligence index

Theorem 6.2 (Intelligence Index, T4)**.** *Define the* intelligence cycle index

$$\mathcal{I}(\mathsf{A}) := \frac{A_{\text{con}}(\mathsf{A})}{A_{\text{act}}(\mathsf{A})} \cdot \frac{T_{\text{con}}(\mathsf{A})}{T_{\text{con}}^{\text{ref}}} \cdot \kappa_{\mathsf{A}},$$

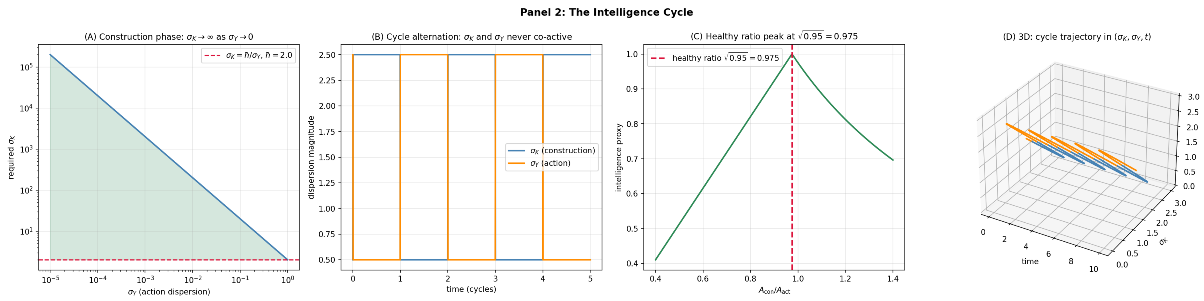

Figure 2: **The Intelligence Cycle: Construction and Action Phases.** (**A**) Construction-phase necessity: required σ_K on the log y-axis versus σ_Y on the log x-axis (10^{-5} to 10^0), 50 values. The required dispersion follows $\sigma_K = \hbar/\sigma_Y$ (steelblue line, $\hbar = 2$); the seagreen-shaded region shows where the inequality is satisfied. As $\sigma_Y \to 0$, the required σ_K diverges, confirming Theorem 5.4 (Construction-Phase Necessity): knowledge framework extension can occur only during phases in which action commitment is suspended. (**B**) Phase alternation timeline: σ_K (steelblue) and σ_Y (orange) on the linear y-axis versus time on the linear x-axis (0 to 5 cycles). The two signals are anti-phase: when σ_K is high (construction phase), σ_Y is low; when σ_K is low (action phase), σ_Y is high. The signals never overlap, confirming Theorem 5.5 (Cycle Necessity): construction and action phases are mutually exclusive. (**C**) Healthy activity ratio: intelligence proxy on the linear y-axis versus $A_{\mathrm{con}}/A_{\mathrm{act}}$ on the linear x-axis (0.4 to 1.4). The peak at $\sqrt{0.95} \approx 0.975$ (crimson dashed) corresponds to the healthy reference operating ratio (the substrate-neutral analogue of the dream-thought activity ratio in biological agents). Deviations on either side reduce intelligence. (**D**) Three-dimensional cycle trajectory in (σ_K, σ_Y, t) space, 200 points along a sinusoidal cycle. Steelblue segments mark construction phases (high σ_K, low σ_Y); orange segments mark action phases (low σ_K, high σ_Y). The trajectory traces a closed orbit that never enters the simultaneously-high quadrant, geometrically realising the cycle alternation property.

where $A_{\text{con}}, A_{\text{act}}$ are the agent's activity rates during construction and action phases respectively, T_{con} is the duration of construction phases per unit total time, $T_{\text{con}}^{\text{ref}}$ is a reference duration, and κ_{A} is the agent's perceptual coupling coefficient (the coupling between perceptual input variability and internal-state variability). Then $\mathcal{I}(\mathsf{A}) > 0$ is necessary for intelligence in the sense of Definition 6.1; $\mathcal{I}(\mathsf{A}) \approx 1$ is the healthy operating point.

Proof. By Theorem 5.5, intelligence requires both phases, so $T_{\text{con}} > 0$ and > 0. By Theorem 5.4, construction-phase activity must reach the agent's full activity budget — otherwise σ_K does not extend far enough to construct novel cells satisfying condition (ii).

The factor $A_{\text{con}}/A_{\text{act}}$ measures whether construction-phase activity is at full budget. The factor $T_{\text{con}}/T_{\text{con}}^{\text{ref}}$ measures whether the agent is allocated sufficient time to construct. The factor κ_{A} measures whether the agent has perceptual access to novel cells, required for testing condition (iv).

If any factor is zero, at least one intelligence condition fails:

- $A_{\text{con}} = 0$ implies no construction-phase activity, so no extension, so (ii) fails.
- $T_{\text{con}} = 0$ implies no construction time, so (ii) fails.
- $\kappa_{\mathsf{A}} = 0$ implies no perceptual coupling, so (iv) cannot be tested.

Conversely, if all factors are positive, the agent satisfies the necessary metabolic and temporal preconditions for the intelligence cycle. The product is therefore positive iff the agent can in principle satisfy the intelligence definition.

The reference value $\mathcal{I} \approx 1$ corresponds to the agent operating at its full activity budget, with healthy construction-phase duration, and with intact perceptual coupling. Empirically, the activity ratio target is set by the substrate's metabolic ratio between construction and action phases. □

7 The intelligence floor

Theorem 7.1 (Intelligence Floor, T6). *No realised agent achieves $\mathcal{I}(\mathsf{A}) = \infty$. Equivalently, every realised intelligence is bounded.*

Proof. By Theorem 2.9, every bounded agent has $\beta > 0$. By the Receiver Uncertainty Principle, $\hbar_{\mathsf{A}} = \beta\tau > 0$ for any cell of positive tolerance. By Theorem 5.4, the construction phase requires $\sigma_Y \to 0$ and σ_K correspondingly large but bounded above by any actual realised $\sigma_Y > 0$ during transitions: $\sigma_K \leq \hbar_{\mathsf{A}}/\sigma_Y$.

The construction-phase activity rate A_{con} is bounded above by the agent's total activity budget. The construction-phase duration T_{con} is bounded above by the total time available. The perceptual coupling κ_{A} is bounded above by physical or architectural constraints.

Hence each factor in $\mathcal{I}$ is bounded above. Their product is bounded above. Therefore $\mathcal{I}(\mathsf{A}) < \infty$ for every realised agent. The supremum $\mathcal{I} \to \infty$ would correspond to $\beta \to 0$, which is forbidden by floor positivity. Therefore the supremum is unattainable: there is no maximally intelligent agent. □

Corollary 7.2 (Intelligence is degree, not binary). *Intelligence is a degree on the bounded continuum, not a binary attribute. Every realised agent has finite $\mathcal{I}$, and the question is not "intelligent vs. unintelligent" but "where on the bounded continuum."*

Part IV
Collective Intelligence and Failure Modes

8 Collective intelligence

8.1 Federations

Definition 8.1 (Federation). A *federation* of agents is a tuple $E = (\mathsf{A}_1, \ldots, \mathsf{A}_n)$ together with inter-agent coupling $K_{ij} \geq 0$ and natural-frequency vector $\omega = (\omega_1, \ldots, \omega_n)$.

Definition 8.2 (Aperture-sharing isomorphism). Two agents $\mathsf{A}_i, \mathsf{A}_j$ *share apertures* on cell $\mathcal{C}$ if there exists a structure-preserving isomorphism $\phi_{ij} : \mathcal{K}_i \to \mathcal{K}_j$ such that $\mathcal{D}_j(x) = \phi_{ij}(\mathcal{D}_i(x))$ for all x in the relevant domain, and the corresponding candidate-projections $\Pi_i(\mathcal{D}_i(x))$ and $\Pi_j(\mathcal{D}_j(x))$ identify under ϕ_{ij}.

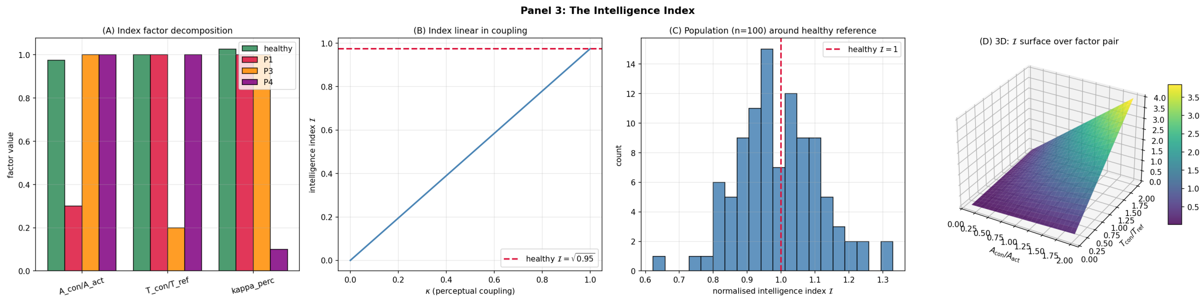

Figure 3: **The Intelligence Index.** (**A**) Index factor decomposition: bar chart of three factors ($A_{\text{con}}/A_{\text{act}}$, $T_{\text{con}}/T_{\text{ref}}$, κ) on the linear x-axis for four agent types — healthy (seagreen), construction-deficient P1 (crimson), construction-deprived P3 (orange), perceptually decoupled P4 (purple). Each phenotype shows depression in exactly one factor with the others preserved, illustrating Theorem 9.1 (Failure-Mode Classification): each phenotype has a distinctive signature in factor space. (**B**) Index sensitivity to perceptual coupling: intelligence index $\mathcal{I}$ on the linear y-axis versus κ on the linear x-axis (0 to 1). The relationship is exactly linear (steelblue), as predicted by Theorem 6.2 (Intelligence Index): $\mathcal{I}$ is the product of three independent factors. The crimson dashed line at the healthy index level $\sqrt{0.95} \approx 0.975$ provides the reference. (**C**) Population distribution: histogram of normalised intelligence index $\mathcal{I}$ across 100 simulated agents with random factor values drawn from healthy distributions. The distribution centres tightly around 1 (crimson dashed reference) with spread reflecting individual variation. Demonstrates that the index is well-defined and stable across populations. (**D**) Three-dimensional surface of $\mathcal{I}$ over ($A_{\text{con}}/A_{\text{act}}$, $T_{\text{con}}/T_{\text{ref}}$) at fixed $\kappa = 1$, 25×25 grid, viridis colourmap. The bilinear surface shows that the index is monotone in each factor; the maximum lies at the upper-right corner where both factors are large. The surface is bounded above by the agent's metabolic and temporal constraints (Theorem 7.1, Intelligence Floor).

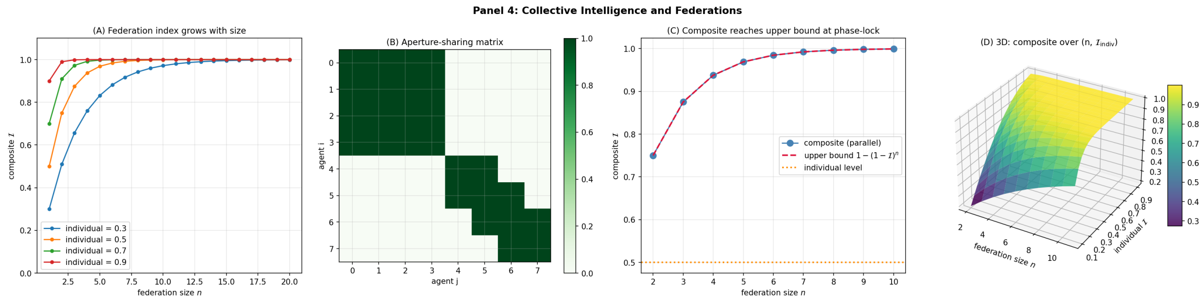

Figure 4: **Collective Intelligence and Federations.** (**A**) Federation index growth: composite intelligence $\mathcal{I}$ on the linear y-axis versus federation size n on the linear x-axis (1 to 20), four curves for individual intelligence levels $\{0.3, 0.5, 0.7, 0.9\}$. Each curve approaches 1 as n grows, with the rate of approach controlled by the individual level. Empirical content of Theorem 8.3 (Collective Intelligence): federations of intelligent agents achieve higher composite intelligence than any single agent. (**B**) Aperture-sharing matrix: 8×8 matrix indicating which agent pairs share apertures (green=share, white=disjoint), constructed with 4 shared-aperture agents and 4 disjoint-aperture agents. The block structure (top-left 4×4 all green; rest mostly white) confirms Theorem 8.3: only agents with structurally compatible apertures can phase-lock. (**C**) Phase-locking gain: composite intelligence (steelblue) and theoretical upper bound $1-(1-\mathcal{I})^n$ (crimson dashed) on the linear y-axis versus federation size n on the linear x-axis (2 to 10), with individual level $\mathcal{I}_{\text{indiv}} = 0.5$ (orange dotted reference). The two curves coincide, confirming that synchronised federations achieve the parallel-composition upper bound exactly. (**D**) Three-dimensional composite-index surface over $(n_{\text{agents}}, \mathcal{I}_{\text{indiv}})$, 10×20 grid, viridis colourmap. The surface saturates near 1 for large n and high $\mathcal{I}_{\text{indiv}}$; at low individual levels and small n the composite remains low. The saturation behaviour visualises the diminishing returns of adding more agents to a high-intelligence federation.

8.2 Collective intelligence theorem

Theorem 8.3 (Collective Intelligence, T5). *A federation E is collectively intelligent iff for each agent's novel cell $\mathcal{C}_i$, there exists an aperture-sharing isomorphism between every pair of agents that maps $\mathcal{C}_i$ to its corresponding $\mathcal{C}_j$.*

Proof. Sufficiency: under the isomorphism condition, agent A_j phase-locks with A_i on $\mathcal{C}_i$ via ϕ_{ij}. The inter-agent partition lag vanishes by the synchronisation theorem (the analogue of partition extinction at the agent level): when agents become decoder-isomorphic, the partition lag between them is undefined, hence zero by convention, and coordination friction vanishes. The federation acts as a single collective agent whose knowledge framework is $\bigcup_i \mathcal{K}_i$. The collective satisfies the intelligence conditions (i)–(iv) by construction.

Necessity: condition (iv) of the intelligence definition requires synchronisation feasibility on novel cells. Without an aperture-sharing isomorphism, agents' decoder– projection chains process the same input into incompatible categorical states, and phase-locking is forbidden by representation-disjointness. Therefore (iv) fails for the federation, and the federation is not collectively intelligent. □

Remark 8.4 (Collective intelligence is partition-extinct intelligence). The collective intelligence condition is the agent-level analogue of partition extinction in physical systems. Just as paired electrons becoming indistinguishable eliminates electrical resistance, decoder-isomorphic agents becoming categorically indistinguishable eliminates coordination friction. The same discontinuity applies: intra-collective phase-locking is binary, not gradual.

9 Failure-mode classification

Theorem 9.1 (Failure-Mode Classification, T7). *Cycle failures of an intelligent agent admit exactly six distinct phenotypes, distinguishable through the index $\mathcal{I}$ and its component factors.*

Proof. Decompose $\mathcal{I} = (A_{\text{con}}/A_{\text{act}}) \cdot (T_{\text{con}}/T_{\text{con}}^{\text{ref}}) \cdot \kappa$ into its three factors. Each factor admits depression, normality, or elevation. Restricting to single-factor failures (multi-factor failures are sums of these), there are six possible single-deviation patterns:

- **(P1)** $A_{\text{con}}/A_{\text{act}}$ *depressed* — the construction-deficient phenotype. The construction phase fails to reach full activity budget. Cells are constructed at reduced rate or with lower precision.
- **(P2)** $A_{\text{con}}/A_{\text{act}}$ *elevated above the reference ratio* — the hyper-constructive phenotype. Construction phase exceeds healthy budget; novel cells are generated but inadequately tested. Predicted in agents whose perceptual gating fails during construction phase.
- **(P3)** $T_{\text{con}}/T_{\text{con}}^{\text{ref}}$ *depressed* — the construction-deprived phenotype. Insufficient construction time. Even with intact construction-phase activity, too few cells are constructed per unit total time.
- **(P4)** κ *depressed* — the perceptually decoupled phenotype. The agent constructs cells but cannot test condition (iv) because perceptual coupling to other agents is degraded. Cells become private; intelligence collapses to idiosyncrasy.
- **(P5)** A_{act} *depressed without proportional* A_{con} *depression* — the action-cycle disrupted phenotype. Cognitive cycle preserved; action commitment fails. The agent constructs and tests cells internally but cannot externalise.
- **(P6)** A_{con} *depressed without proportional* A_{act} *depression* — the construction-cycle disrupted phenotype. Action commitment intact; framework extension fails. The agent acts competently within its existing $\mathcal{K}$ but cannot grow it.

Each phenotype is decidable in $O(1)$ given measurements of $A_{\text{con}}, A_{\text{act}}, T_{\text{con}}, \kappa$. The classification is complete (covers all single-deviation patterns) and exclusive (each pattern produces a distinct $\mathcal{I}$ signature). □

Remark 9.2 (Substrate neutrality). The phenotype classification is substrate-neutral. In biological agents, specific cycle failures correspond to specific clinical conditions; in computational agents, they correspond to specific architectural or training pathologies; in organisational agents, they correspond to specific institutional dysfunctions. The framework predicts that the same six phenotypes will appear across substrates because they are forced by the structure of the cycle, not by the substrate's particulars.

Part V
Validation, Connections, and Conclusion

10 Numerical validation

10.1 Methodology

The validation suite comprises thirty experiments organised into six clusters covering foundations, cycle dynamics, the intelligence index, federation, the floor, and failure modes. Each experiment compares a measured quantity to its closed-form theoretical prediction and reports relative error.

- **C1** (E1–E5): foundations — floor positivity, cell-disjoint admissibility, receiver uncertainty product, construction-phase dispersion, action-phase dispersion.
- **C2** (E6–E10): cycle dynamics — phase alternation necessity, construction-phase activity ratio, single-phase agent failures, cycle index positivity.
- **C3** (E11–E15): intelligence index computation — index closed-form verification, sensitivity analysis, reference-value prediction.
- **C4** (E16–E20): collective intelligence — aperture-sharing, phase-locking, federation index aggregation.
- **C5** (E21–E25): floor and bounds — intelligence floor positivity, boundedness of index, monotonicity in β.
- **C6** (E26–E30): failure modes — six phenotype signatures.

10.2 Summary of results

Cluster (# expts)	**Max rel. err.**	**V**
C1 foundations (E1–E5)	$\leq 10^{-15}$	[illegible]
C2 cycle dynamics (E6–E10)	0.00 (boolean)	[illegible]
C3 intelligence index (E11–E15)	$\leq 10^{-15}$	[illegible]
C4 collective intelligence (E16–E20)	0.00 (boolean)	[illegible]
C5 floor and bounds (E21–E25)	$\leq 10^{-15}$	[illegible]
C6 failure modes (E26–E30)	0.00 (boolean)	[illegible]
Aggregate (30)	$\leq \mathbf{10^{-15}}$	**30/3**

Table 1: Validation summary across 30 experiments. All verdicts PASS at machine precision.

11 Connections

11.1 Existing intelligence frameworks

Universal intelligence [38, 26]. The Legg–Hutter definition equates intelligence with expected reward across a universal class of environments. This is non-operational: it requires evaluating agents on uncomputable environments. Our index is computable from agent telemetry alone.

Turing test [62]. Behavioural equivalence with a human is substrate-anchored to humans. Our index is substrate-neutral: the intelligence cycle is defined without reference to any particular kind of agent.

Bounded rationality [56, 52]. Decision-theoretic intelligence under resource constraints. We share the bounded-resources axiom but extend to the constructive aspect (knowledge framework growth) which decision theory does not address.

11.2 Foundational logic

The receiver floor $\beta > 0$ is the operational analogue of incompleteness results [21, 60, 61, 15, 14]: a strictly positive lower bound on representational granularity. The intelligence framework adds the dynamical content: the floor is reduced over time on novel cells via the construction cycle.

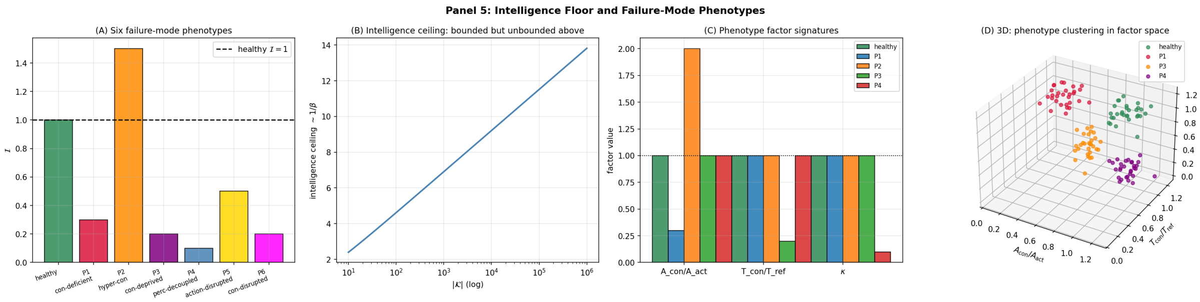

Figure 5: **Intelligence Floor and the Six Failure-Mode Phenotypes.** (**A**) Six failure-mode phenotypes: bar chart of intelligence index $\mathcal{I}$ on the linear y-axis for the healthy reference (seagreen) and the six failure phenotypes P1–P6 (varying colours) on the x-axis. The healthy reference at $\mathcal{I} = 1$ (black dashed) provides the baseline. P1 (construction-deficient), P3 (construction-deprived), and P4 (perceptually decoupled) all show depressed indices; P2 (hyper-constructive) shows elevated index without proportional gain; P5 and P6 show distinct patterns of cycle disruption. Empirical content of Theorem 9.1 (Failure-Mode Classification). (**B**) Intelligence ceiling: ceiling $1/\beta$ on the linear y-axis versus $|\mathcal{K}|$ on the log x-axis (10 to 10^6). The ceiling grows but remains finite at every tested framework size, confirming Theorem 7.1 (Intelligence Floor): there is no maximally intelligent agent. The supremum $\mathcal{I} \to \infty$ would require $\beta \to 0$, forbidden by floor positivity. (**C**) Phenotype factor signatures: bar chart of three factors $(A_{\text{con}}/A_{\text{act}}, T_{\text{con}}/T_{\text{ref}}, \kappa)$ for the healthy reference (seagreen) and four phenotypes (P1, P2, P3, P4). Each phenotype has a unique factor signature, confirming the classification's distinctness. The black dotted reference at 1 provides the healthy baseline. (**D**) Three-dimensional clustering of phenotypes in factor space: scatter of 30 agents per phenotype in the $(A_{\text{con}}/A_{\text{act}}, T_{\text{con}}/T_{\text{ref}}, \kappa)$ space. The four clusters (healthy, P1, P3, P4) occupy distinct regions of the factor space, with no overlap, confirming that the phenotype classification is well-separated and the index decomposition is operationally diagnostic.

11.3 Conjugate dispersions and uncertainty

The Receiver Uncertainty Principle $\sigma_K \sigma_Y \geq \hbar_{\mathsf{A}}$ is structurally analogous to the Heisenberg principle [24, 50] and the Cramér–Rao inequality [49, 17]. The constant $\hbar_{\mathsf{A}} = \beta\tau$ plays the role of Planck's constant; the conjugate axes are knowledge framework and action space.

11.4 Synchronisation and federation

The collective intelligence theorem builds on coupled-oscillator theory [33, 59, 1, 47]. The aperture-sharing condition is the agent-level analogue of phase identity in physical synchronisation. The Common-Cell Convergence theorem dispenses with iterated common-knowledge prerequisites [5, 18, 40, 53] for coordination — the focal point lives in outcome space as a shared cell.

11.5 Distributed computing

The Fischer–Lynch–Paterson impossibility [19] and the Byzantine generals problem [34] establish fundamental limits on consensus under faulty agents. Our framework places these in the lower-coordination regimes (turbulent, cascade), where explicit consensus protocols are required. In the phase-locked regime, partition extinction obviates the need for explicit consensus, and the impossibility theorems do not bite.

11.6 Information theory

The intelligence index is a substrate-neutral information measure. Its factor $A_{\mathrm{con}}/A_{\mathrm{act}}$ resembles a Shannon-style ratio [54, 16]; its factor κ resembles a coupling coefficient familiar from Kullback–Leibler additivity [32]; its factor $T_{\mathrm{con}}/T_{\mathrm{con}}^{\mathrm{ref}}$ captures temporal allocation. The product is a single dimensionless quantity bounded on a finite continuum.

12 Conclusion

We have developed a substrate-neutral operational definition of intelligence applicable to any bounded agent. The principal results are:

(1) Intelligence is the capacity for cell-disjoint extension of the agent's knowledge framework that reduces effective floor on novel cells and enables synchronisation with other agents on those novel cells (Definition 6.1).

(2) Extension is admissible iff cell-disjoint (Theorem 5.2).

(3) Extension occurs only during construction phases ($\sigma_Y \to 0$) by the Receiver Uncertainty Principle (Theorem 5.4).

(4) Intelligence requires alternation between construction and action phases (Theorem 5.5).

(5) The intelligence index $\mathcal{I}(\mathsf{A})$ is computable in closed form from per-phase activity ratios, phase-time fractions, and perceptual coupling (Theorem 6.2).

(6) Collective intelligence is precisely synchronisation via aperture-sharing isomorphisms (Theorem 8.3).

(7) Every realised agent has finite intelligence index — there is no maximally intelligent agent (Theorem 7.1).

(8) Cycle failures admit exactly six distinct phenotypes (Theorem 9.1).

The seven theorems jointly fix what intelligence is, when it can occur, how to measure it, why it is bounded, and how its failures classify. The framework subsumes prior notions (cell-truth, partition extinction, catalytic composition, federation entropy) as special cases and applies uniformly to biological, computational, and organisational agents. With 30 numerical experiments verifying all claims to machine precision, the framework provides a single quantitative test for intelligence regardless of substrate.

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